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Marine pollutant exposures and human milk extracellular vesicle-microRNAs in a mother-infant cohort from the Faroe Islands

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ABSTRACT

Background/Aims: Early life exposures to marine contaminants can adversely impact child health but modes of action are unclear. Human milk contains extracellular vesicles (EVs) that can transport biologically relevant cargo from mother to infant, including microRNAs (miRNAs), and may partly mediate the effects of pollutants on child health. However, the role of marine pollutants on miRNA expression in milk EVs is unexplored.

Methods: We isolated EV RNA from 333 milk samples collected between 2 and 74 days postpartum from a Faroese birth cohort born 1997–2000 and sequenced 2083 miRNAs using a targeted library preparation method. We quantified five perfluoroalkyl substances (PFAS), pesticide metabolite *p,p'*-dichlorodiphenyldichloroethylene (DDE), and the sum of three major polychlorinated biphenyls (ΣPCBs) in maternal serum at 34 weeks of gestation and maternal hair total mercury (Hg) at birth. We used negative binomial regressions to estimate associations between individual pollutants and 418 reliably expressed EV-miRNAs adjusted for potential confounders. We performed sparse principal components (PCs) analysis to derive the first four components of the EV-miRNA data and examined associations between pollutants and PCs using Bayesian kernel machine regression (BKMR).

Results: We observed no associations between pollutants and individual EV-miRNA expression after controlling the false discovery rate at 0.1. However, BKMR suggested that Hg was positively associated with PC1 and negatively associated with PC3, while ΣPCBs was negatively associated with PC3, and two PFAS were associated with PC4. Exploration of PC loadings followed by pathway analyses suggested that miRNAs in PC1 (miR-200b-3p, miR-664a-3p, miR-6738-5p, miR-429, miR-1236-5p, miR-4464, and miR-30b-5p) may be related to Hg neurotoxicity, while remaining PCs require further research.

Conclusions: Our findings suggest that groups of milk EV-miRNAs may better serve as environmental biomarkers than individual miRNAs. Future studies are needed to elucidate the role of milk EV-miRNAs in child health following prenatal exposures.

1. Introduction

Exposures to environmental pollutants during pregnancy or in early postnatal development can have adverse consequences for both the mother and the developing child. Marine contaminants, such as

perfluoroalkyl substances (PFAS), *p,p'*-dichlorodiphenyltrichloroethane (DDT), polychlorinated biphenyls (PCBs), and methylmercury (MeHg), are produced by or mobilized from human industrial and agricultural processes. Due to their lipophilic nature or protein binding capacity, many of these compounds partition into human milk, where they can

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modify maternal lactation and transfer to the breastfeeding infant (Fängström et al., 2005; Grandjean et al., 1995; Mogensen et al., 2015; Timmermann et al., 2017).

Breastfeeding plays a central role in the long-term programming of infant health, providing critical nutrients to developing newborns, including essential vitamins and minerals, proteins, carbohydrates, and lipids (Dror and Allen, 2018). In this capacity, human milk can influence the maturation and development of several infant organ systems (Gartner et al., 2005). Recent research in experimental models suggests that in addition to other bioactive compounds, human milk can transfer bioactive nucleic acids contained within extracellular vesicles (EVs) from mother to child (de la Torre Gomez et al., 2018). EVs are nano-sized, lipid-bound vesicles that are found in all human biofluids and transport biological materials from donor to recipient cells (Bobrie et al., 2012; Godoy et al., 2018). Released into human milk by the luminal epithelium, *in vitro* evidence suggests that milk EVs can survive acidic conditions mimicking the stomach environment and can alter gene expression in recipient cells (Jiang et al., 2021). Furthermore, *in vivo* research on milk EVs suggests that they may induce systemic metabolic alterations in offspring (Baier et al., 2014; Jiang et al., 2021).

EVs contain a high content of microRNAs (miRNAs), small non-coding RNAs that are 21–22 nucleotides in length (Valadi et al., 2007). miRNAs function in the repression of messenger RNA (mRNA) transcripts, either via enhancement of mRNA degradation or obstruction of mRNA translation, and play a critical role in development, disease, and the cellular stress response (Bartel, 2004). Previous research demonstrates that milk miRNA content is associated with maternal characteristics, such as body mass index (BMI), Type I Diabetes, diet, stress, smoking, or premature delivery (Jiang et al., 2021; Kupsco et al., 2021; Shah et al., 2021; Zamanillo et al., 2019). Furthermore, EV-miRNA content in blood is sensitive to environmental exposures in both experimental and human population studies (Qin et al., 2020). This suggests that human milk EV-miRNA content might be altered following pollutant exposure during pregnancy, thereby transferring important biological signals to the developing infant. However, to date, no study has investigated the associations of environmental toxicant exposures on human milk EV-miRNA expression. Using a population of mother-child pairs from a North Atlantic fishing community, our goal was to examine milk EV-miRNA as biomarkers of prenatal marine pollutant exposure. We specifically aimed to leverage mixture methods to disentangle impacts of highly correlated pollutants and investigate associations of pollutants on joint miRNA expression levels.

2. Materials and methods

2.1. Study population

We leveraged information and biospecimens from women from the Faroe Islands recruited during pregnancy between 1997 and 2000 as part of a population-based prospective cohort study (Weihe and Grandjean, 2012). The cohort enrolled 656 mother-child pairs with singleton births at 34 weeks of gestation at the National Hospital in Tórshavn, Faroe Islands. Included births represented approximately 60% of all pregnancies in Tórshavn during the study period with most attrition due to work schedules in the busy ward or scheduling conflicts. The study was approved by the Faroese ethical review committee and the Institutional Review Boards at all participating institutions (Harvard T.H. Chan School of Public Health and Columbia University Mailman School of Public Health), and all women provided written informed consent at each visit.

2.2. Environmental exposure assessment

We collected maternal serum at 34 weeks of gestation for persistent organic pollutant assessment. Maternal hair was collected at parturition for mercury (Hg) analysis. Serum organochlorides (major PCB congeners

CB-138, CB-153, and CB-180, DDT, and *p,p'*-dichlorodiphenyldichloroethylene [DDE]) underwent solid-phase extraction and were quantified using gas chromatography (GC) with electron capture detection as previously described (Grandjean et al., 2012). The major PCB congeners represented approximately 50% of the total PCB burden. To avoid issues with values below the limit of detection for minor congeners, we calculated a simplified Σ PCB concentration as the sum of major congeners CB-138, CB-153, and CB-180 multiplied by 2 (Grandjean et al., 1995). None of the Σ PCB or DDE values were below the limit of detection (LOD) of 0.03 ng/mL. As PCBs and DDE are highly lipophilic, values were adjusted by the total serum lipid (triglycerides + cholesterol) concentration determined using the Cypress Diagnostics kit (Langdorp, Belgium) and are expressed in $\mu\text{g/g}$ lipid.

Serum PFAS concentrations were quantified using high-pressure liquid chromatography (HPLC) coupled to tandem mass spectrometry (MS/MS) (Vorkamp et al., 2014). Quantified PFASs included: perfluorooctane sulfonate (PFOS), perfluorooctanoic acid (PFOA), perfluorohexane sulfonic acid (PFHxS), perfluorodecanoic acid (PFDA), and perfluorononanoic acid (PFNA). PFOS concentrations were calculated as a sum of linear and branched isoforms. All samples were quantified above the LOD (>0.03 ng/mL) and are expressed in ng/mL.

Total hair Hg was measured in a 1-cm hair segment cut proximal to the scalp as previously described (Grandjean and Budtz-Jørgensen, 2007). This section primarily reflects total methylmercury exposure during the 2nd and 3rd trimesters (Grandjean et al., 1999). Mercury levels are presented as $\mu\text{g/g}$ hair, and values were above the LOD.

2.3. Human milk collection

Milk was collected as previously described (Kupsco et al., 2021), either by hand expression or by pump at clinical follow-up visit that was scheduled two weeks after the expected delivery date, as assessed by routine ultrasound. Milk was frozen at -80°C until analysis. Out of the 656 initially enrolled, 364 mothers (55%) had available milk samples.

2.4. Human milk EV-RNA isolation

Milk EV-RNA was isolated from 1 to 2 mL of skim milk as previously described (Kupsco et al., 2021). Studies have shown that the milk lipid fraction contains distinct miRNA content in comparison to the skim fraction (Alsaweed et al., 2016a). We chose to focus on skim milk rather than the whole milk to ensure that differential milk fat content between subjects did not influence our results. Additionally, lipoproteins and lipids are known to form vesicles that may be co-isolated with EVs (Vickers et al., 2011), thus contaminating the EV population. To ensure we were only capturing EVs that were intentionally secreted by the cell, we removed the lipid layer from samples prior to EV isolation. In the present analysis, we considered exosomes (30–150 nm) and microvesicles (50–1000 nm) as EVs, as they are actively released from live cells, and excluded apoptotic bodies (>1000 nm). In brief, samples were centrifuged at 200g at 4°C for 10 min followed by 3000g at 4°C for 15 min, then isolated with the ExoEasy Maxi Kit (Qiagen, Germantown, MD), a membrane-affinity based method, following the manufacturer's instructions. EV presence and enrichment were previously confirmed (Kupsco et al., 2021). RNA was isolated from EVs with the miRNeasy Serum/Plasma Maxi kit (Qiagen, Germantown, MD), then cleaned using the RNA Clean & Concentrator-5 Kit (Zymo Research, Irvine, CA). Sample integrity was qualitatively examined with the TapeStation 4200 total RNA ScreenTape chips (Agilent, Santa Clara, CA) prior to sequencing. EV quality assessments are available on the EV-TRACK knowledgebase (EV-TRACK ID: EV200197).

2.5. Analysis of human milk EV-miRNA by targeted sequencing

Milk EV-miRNAs were sequenced as previously described (Kupsco et al., 2021). Libraries were prepared at the University of North Carolina

High Throughput Sequencing Core, with 30 ng of RNA using the Edge-Seq miRNA Whole Transcriptome Assay (HTG Molecular Diagnostics, Inc., Tucson, AZ), which targets 2083 known mature miRNAs. Samples were barcoded and multiplexed so that 90 samples, three duplicate internal human milk EV-RNA controls, and three internal human brain tissue controls were sequenced per run, with four sequencing runs in total. Sequencing data preprocessing was performed at HTG Molecular Diagnostics in Tucson, AZ. All samples passed quality control, including percentage of bases with a quality score greater than 30 (Q30 score) $\geq$ 85%, percentage of clusters passing filter $\geq$ 75%, and $180 < CD < 290$. However, one library had low read counts $< 100,000$ and was removed from the analysis. We observed high reproducibility between samples, with correlations between human milk EV triplicate samples ranging from 0.87 to 0.95 across the four sequencing runs and known internal brain control replicate sample correlations ranging between 0.94 and 0.97. The sequencing datasets analyzed in the current study are available on Gene Expression Omnibus (GEO) with accession number: GSE146880.

2.6. Information on covariates

The Faroe Islands are a North Atlantic fishing community which is fairly homogeneous in regard to socioeconomic and lifestyle factors. Information on maternal age at delivery (as a continuous variable), child sex (male or female), and gestational age at birth (continuous), parity (multiparous or primiparous), pre-pregnancy BMI (continuous), gestational weight gain (continuous), smoking during pregnancy (yes or no), maternal education ($<$ high school, high school, or $>$ high school), gestational diabetic status (yes or no), and preeclampsia (yes or no) was extracted from obstetric records. Information on fish consumption during pregnancy was assessed by questionnaire as servings per week in a subset of participants ($n = 168$) and was dichotomized into ≤ 2 servings or > 2 servings per week for sensitivity analyses.

2.7. Statistical analysis

There were 333 individuals with overlapping EV-miRNA and exposure data. Exposures were log2 transformed prior to analyses to handle right-skewed distributions. To reduce the dimensionality of the EV-miRNA sequencing data and retain only EV-miRNAs with high confidence, we filtered data to include only EV-miRNAs with log2 counts per million (CPM) > 4 , as technical replicates indicated high reproducibility among these miRNAs (Pearson's correlations 0.9–0.99). This resulted in 418 EV-miRNAs for analysis.

Associations between EV-miRNAs and exposures were assessed using negative binomial regression, which accommodates overdispersion in the counts relative to a Poisson assumption. Normalized library sizes were calculated with the *edgeR* upper quartile normalization method (Bullard et al., 2010; Robinson et al., 2010) and included as the offset in the negative binomial models. Each EV-miRNA and exposure were assessed individually and adjusted for EV-miRNA precision variables (i. e., RNA processing batch, RNA concentration, milk collection day post-delivery) and confounders, which were identified as potential causal factors of both exposure and outcome using directed acyclic graphs (maternal education [$<$ high school, high school, $>$ high school], maternal age, parity [primi- or multiparous], and smoking during pregnancy [yes or no]) (Supplemental Fig. S1). Maternal BMI, gestational weight gain, preeclampsia, gestational diabetes, and gestational age at birth, were all considered to lie along the potential causal pathway and were not included as covariates. To ensure we retained true biological variation while minimizing the undue influence by EV-miRNA outliers, for each EV-miRNA model, individuals with studentized residuals > 5 were excluded from the analysis. This resulted in 356 models with the full population of 333, 51 models with 332 participants, 9 models with 331 participants and 2 models with 330 participants. Model coefficients were exponentiated to yield estimates of relative rate,

representing the fold change in mean EV-miRNA abundance per doubling of exposure. P-values for associations of all EV-miRNAs with individual pollutants were adjusted for multiple comparisons by controlling the False Discover Rate (FDR) (Benjamini and Hochberg, 1995), with significance set at $FDR < 0.1$.

To assess associations of pollutants with total EV-miRNA variability while reducing the dimensionality of the EV-miRNA data, we employed sparse principal components analysis (SPCA) with the *elasticnet* package in R (Zou et al., 2006). Using EV-miRNA log2CPMs batch corrected with Combat (Leek et al., 2012), we first examined elbow plots of the percent variation explained by each unpenalized PCA to select four principal components (PCs) as the optimal number for investigation (Supplemental Fig. S2). We then tested a range of L1 (lasso) penalty parameters (25–200 in increments of 25) and selected 75 as a penalty parameter that retained adequate biological variation while remaining interpretable (Supplemental Figs. S3 and S4). SPC loadings for the first 4 PCs were clustered using hierarchical clustering with Euclidean distance and complete linkage into 8 clusters.

Finally, as marine contaminant exposures occur in a highly correlated complex mixture, we assessed associations between the pollutants and each PC using Bayesian Kernel Machine Regression (BKMR) (Bobb et al., 2015). BKMR uses a kernel function to flexibly model highly correlated exposures without prior specification of the exposure-response function (Bobb et al., 2015; Valeri et al., 2017). We calculated z-scores for all PCs and log2 transformed pollutants prior to analyses to enhance comparisons of associations between different pairs of PCs and pollutants. We ran 100,000 iterations of a Markov chain Monte Carlo algorithm with a hierarchical variable selection of exposures grouped into PFASs and non-PFAS exposures based on the exposure correlation structures with the following model: $Y_i = h([Group_1 = (PFOS, PFOA, PFHxS, PFNA, PFDA), Group_2 = (SPCB, DDE, Hg)] \cdot) + \beta^T C_i + \epsilon_i$. A Gaussian kernel function was used to model $h()$ on the outcome Y_i , which allows for potential nonlinear and non-additive exposure-response functions, β^T denotes coefficients for the vector of covariates C_i for the i -th individual, and ϵ_i represents the error. Model convergence and fit were assessed with a visual examination of trace plots for all parameters across the 100,000 iterations. Posterior inclusion probabilities (PIPs) were estimated for each exposure (Supplemental Fig. S5).

2.8. Sensitivity analyses

To assess the robustness of our findings, we performed several sensitivity analyses. As fish consumption is a significant determinant of marine contaminant exposures, and because diet may influence EV-miRNA expression in human milk, we co-adjusted models for maternal fish consumption during pregnancy ($n = 168$). Although we reasoned that maternal BMI and gestational weight gain may be on the causal pathway between pollutants and EV-miRNAs, given the strong associations between maternal weight and miRNAs (Kupsco et al., 2021), we performed additional models co-adjusting for maternal BMI and gestational weight gain. As recent evidence suggests that child sex may also impact milk EV-miRNA expression (Bozack et al., 2021), we considered child sex as a potential precision variable and adjusted for it in our sensitivity analyses. We also removed women who developed preeclampsia ($n = 4$) during pregnancy to ensure this condition was not modifying our results. Many of our participants reported smoking during pregnancy (27.3%), so we repeated our analyses stratified by smokers and nonsmokers. To confirm that our choice of upper quantile normalization was not driving our results, we examined associations using the Trimmed-Mean of M (TMM) normalization method with *edgeR* (Robinson et al., 2010). To also ensure that our removal of outlying values in negative binomial regressions was not vastly different from models containing all individuals, we re-ran models containing all points. We altered our SPCA penalty parameter to 25 and 125 to confirm that our results were robust to penalty specification. Finally, we systematically

varied the prior distribution hyperparameters by 50% greater than the default and 50% less than the default to ensure our results were robust to prior specification.

All analyses were performed in the statistical program R (R Project for Statistical Computing, CRAN, The Comprehensive R Archive Network, Vienna) (R Core Team, 2018).

2.9. Milk EV-miRNA pathway analysis

We performed functional annotation of EV-miRNAs in significant PCs that were also nominally significantly associated with exposures at $p < 0.05$ in individual models. We first used DIANA microT-CDS to predict mRNA targets of miRNAs (Paraskevopoulou et al., 2013; Reczko et al., 2012). DIANA MirPATH version 3 online software (<http://snf-515788.vm.okeanos.grnet.gr/>) (Vlachos et al., 2015) was then used to identify Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways (Kanehisa, 2019; Kanehisa et al., 2020; Kanehisa and Goto, 2000) with significant enrichment ($FDR < 0.05$). To ensure that our results were robust to differing methods of miRNA target prediction, we repeated pathway analyses using TargetScan, which uses conservation of miRNA seed regions for predictions, and Tarbase, a database of experimentally determined interactions (Agarwal et al., 2015, p. 201; Karagkouni et al., 2018; Lewis et al., 2005). We included microT-CDS as our primary analysis as some more recently identified miRNAs were not present in the TargetScan or Tarbase databases.

3. Results

3.1. Participant characteristics

Women included in the analysis were on average 29.4 years of age and were mostly multiparous, with 27.3% reporting smoking during pregnancy (Table 1). Participants included in our analysis ($n = 333$) did not meaningfully differ from the entire cohort ($n = 656$), except they consumed proportionally less fish (Table 1). Milk was collected on average day 15.4 post birth. Of the PFASs, serum PFOS was detected at the highest levels (Table 2). Correlations between pairs of exposures varied by pollutant class. The highest correlations were observed between Σ PCBs and DDE ($r = 0.82$) and between PFDA and PFNA ($r = 0.79$) (Supplemental Fig. S6). The lowest correlations were observed between PFOA and DDE, Σ PCBs, and Hg ($r = 0$ to -0.04) and PFHxS and other pollutants ($r = 0.33$ to 0.08).

Table 1

Characteristics of Faroese women and their pollutant exposures during pregnancy included in this study and the entire Faroes birth cohort.

	Analysis Subset ($n = 333$)	Full Cohort ($n = 656$)
Maternal Characteristics	Mean $\pm$ SD	Mean $\pm$ SD
Age at Parturition (years)	29.4 $\pm$ 4.9	29.2 $\pm$ 5.2
Pre-pregnancy BMI (kg/m ²)	23.8 $\pm$ 3.7	23.9 $\pm$ 4.1
Gestational weight gain (kg)	15 $\pm$ 5.2	14.9 $\pm$ 5.3
Milk Collection Day Postpartum	15.4 $\pm$ 9.7	–
	N (%)	N (%)
Smoked during pregnancy, yes	91 (27.3)	183 (27.9)
Multiparous, yes	258 (77.5)	479 (73.0)
Less than high school education	94 (28.2)	194 (29.6)
High school education	123 (37.0)	247 (37.7)
More than high school education	116 (34.8)	215 (32.8)
Consumed < 2 fish meals per week	108 (64.3)	177 (56.4)
Consumed > 2 fish meals per week	60 (35.7)	137 (43.6)
Gestational Diabetes, yes	30 (9.0)	52 (7.9)
Preeclampsia, yes	4 (1.2)	12 (1.8)

3.2. Associations between marine pollutants and individual milk EV-miRNAs

We observed no FDR significant associations between individual EV-miRNAs and single pollutants in fully adjusted negative binomial models (Fig. 1). There were 15 nominally significant EV-miRNAs associated with DDE, PFOA, and PFDA, 18 associated with PFOS, 21 associated with PFNA, 28 associated with Σ PCBs, 33 associated with PFHxS, and 49 associated with Hg. The strongest association was detected between Hg and miR-128-3p (1.13 fold change in EV-miRNA expression per doubling of Hg, $p = 0.0004$, $FDR p = 0.17$); however, this did not reach statistical significance. No other associations between pollutants and EV-miRNAs were observed.

3.3. Sparse principal components analysis of EV-miRNAs and marine pollutants

Since we hypothesized that pollutant exposures would have stronger associations with sets of EV-miRNAs rather than individual EV-miRNAs, we implemented SPCA to examine total EV-miRNA variability in relation to pollutant exposures. The first 4 PCs explained 23.3% of the total variability in the EV-miRNA data, with PC1 explaining 17% of the variability, PC2 2.2%, PC3 2.8%, and PC4 1.2%. The lower variability explained by PC2 than PC3 is likely induced by the penalization procedure. Across the 4 PCs there were 158 EV-miRNAs from the original 418 with non-zero loadings. Of these, 57 EV-miRNAs with non-zero loadings were included in PC1, 67 were in PC2, 52 were in PC3, and 32 were in PC4. Examination of associations between PCs and pollutants using a mixtures approach found that Hg was positively associated with PC1, Σ PCBs were negatively associated with PC2, Hg was negatively associated with PC3, and PFOS and PFNA were associated with PC4 in opposite directions (Fig. 2).

3.4. Examination of milk EV-miRNA SPC loadings

As PCs themselves are not biologically interpretable, we qualitatively examined the SPC loadings to infer the implications of these associations. We used hierarchical clustering of loadings to provide insight into distinct EV-miRNA loading patterns across PCs and aid in the visual interpretation of results. We also only examined miRNAs that overlapped between PCs and the nominally significant associations observed in individual regression models to ensure the robustness of our interpretations. Hierarchical clustering of PC loadings for the 4 PCs identified 8 clusters of EV-miRNA loadings (Fig. 3).

Cluster 1 was the largest cluster with 76 EV-miRNAs, followed by Cluster 4 with 38 EV-miRNAs (Supplemental Table S1). The smallest cluster was Cluster 3 with 3 EV-miRNAs. For PC1, the most positive loadings were observed in Cluster 6 ($n = 6$ non-zero EV-miRNAs) and the most negative loadings were in Cluster 1 ($n = 30$ nonzero EV-miRNAs). Most of the EV-miRNAs in Cluster 1 were previously identified to be strongly associated with milk collection date postpartum (Kupsco et al., 2021); however, 13 EV-miRNAs in PC1 were also nominally associated with Hg in individual EV-miRNA models at $p < 0.05$ (miR-200b-3p, miR-664a-3p, miR-6738-5p, miR-429, miR-1236-5p, miR-4464, and miR-30b-5p). Pathway analyses of these overlapping EV-miRNAs identified 40 FDR-corrected significant pathways. The top enriched non-cancer pathways were axon guidance, mTOR signaling pathway, ErbB signaling pathway, morphine addiction, and AMPK signaling pathway (Fig. 4). Axon guidance and the ErbB signaling pathway were also enriched when miRNA targets were predicted using TargetScan; however, the top enriched pathway was the phosphatidylinositol signaling system (Supplemental Table S2). Furthermore, the top enriched pathways from Tarbase were fatty acid biosynthesis and oocyte meiosis (Supplemental Table S2).

As BKMR models suggested that Σ PCBs were associated with PC2, we examined the overlap between nominally significant EV-miRNAs

Table 2

Concentrations of marine pollutants in maternal serum collected at 32 weeks of gestation and in hair collected at birth (N = 333).

Pollutant	GM $\pm$ GSD	Minimum	25th Percentile	Median	75th Percentile	Maximum
Hair Mercury ($\mu\text{g/g}$ hair)	2.25 $\pm$ 2.31	0.02	1.43	2.28	3.89	29.9
Serum ΣPCB ($\mu\text{g/g}$ lipid) ^a	0.14 $\pm$ 2.15	0.01	0.09	0.15	0.22	1.3
Serum DDE ($\mu\text{g/g}$ lipid)	0.54 $\pm$ 2.3	0.04	0.33	0.54	0.94	8
Serum PFHxS (ng/mL)	3.99 $\pm$ 2.21	0.79	2.07	4.02	7.75	22.5
Serum PFOS (ng/mL)	26.91 $\pm$ 1.33	9.4	23.1	26.8	32.5	53.2
Serum PFOA (ng/mL)	3.06 $\pm$ 1.41	0.82	2.46	3.14	3.88	7.5
Serum PFNA (ng/mL)	0.6 $\pm$ 1.49	0.16	0.46	0.60	0.78	1.62
Serum PFDA (ng/mL)	0.28 $\pm$ 1.53	0.08	0.22	0.28	0.38	0.98

Note: GM: Geometric Mean, GSD: Geometric Standard Deviation

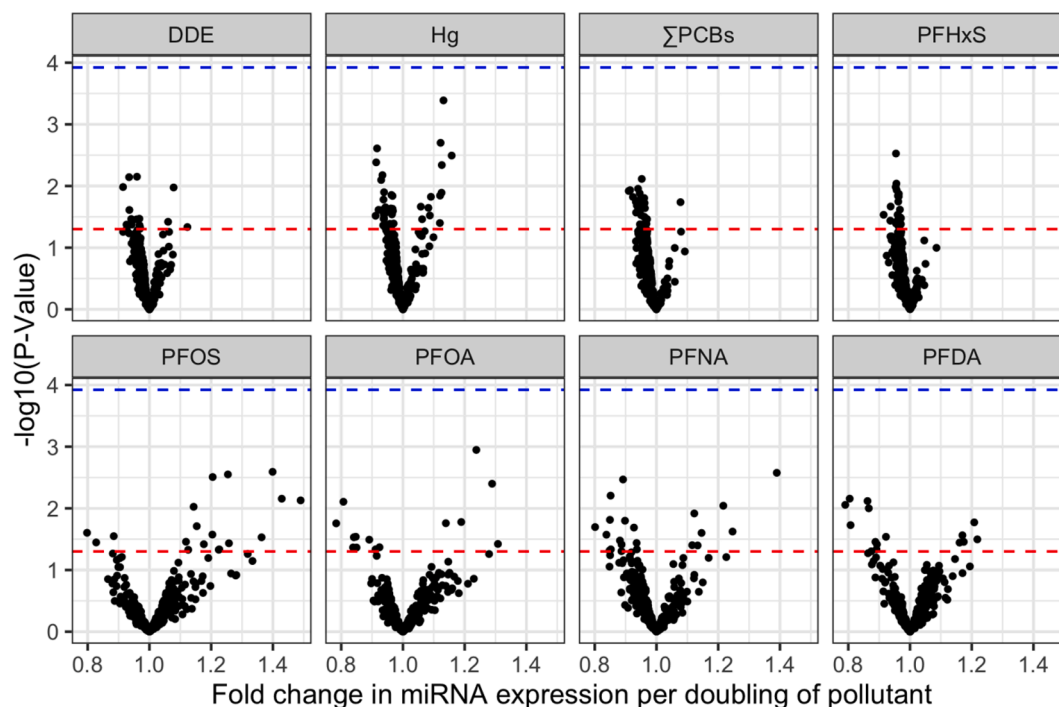
^a The ΣPCB is calculated by summing the concentrations of PCB congeners 138, 153 and 180 multiplied by 2.

Fig. 1. Volcano plots of associations between marine pollutants and milk EV-miRNA expression. Effect estimates expressed as fold change in EV-miRNA expression per doubling of each pollutant. All models adjusted for age at parturition, parity, smoking during pregnancy, education, RNA concentration, RNA batch, and milk collection day postpartum. Dashed red line is equivalent to nominal significance ($p < 0.05$) and dashed blue line is Bonferroni significance ($p = 0.05/418$). No associations reached FDR significance < 0.1 . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

associated with ΣPCBs in individual models and those with non-zero loadings in PC2. This resulted in 3 EV-miRNAs from Clusters 5 and 6 (miR-3907, miR-1913, and miR-4419a). There were six non-cancer KEGG pathways enriched for mRNA targets of these EV-miRNAs, the top of which was thyroid hormone synthesis (Supplemental Fig. S7). No targets of these miRNAs were predicted using TargetScan and Adherens junction was the top pathway from Tarbase, which only identified mRNA targets of miR-1913 (Supplemental Table S3).

There were 7 overlapping EV-miRNAs with non-zero loadings in PC3 and nominal significance in individual EV-miRNA models (miR-492, miR-128-3p, miR-1290, miR-671-3p, miR-92b-5p, miR-3907, miR-4513, miR-6754-5p). The top KEGG pathway enriched for these EV-miRNAs was the ErbB signaling pathway (Supplemental Fig. S8). In contrast, the top pathways predicted by TargetScan and Tarbase were Mucin type O-Glycan biosynthesis and NF-kappa B signaling pathway, respectively (Supplemental Table S4).

Finally, we observed opposing associations between PFOS and PFNA with PC4. The most negative loadings in PC4 were in Cluster 7; although none of these overlapped with nominally significant EV-miRNAs from individual models with PFOS or PFNA. Among the EV-miRNAs with the

most positive loadings in Cluster 4, three were nominally associated with PFOS exposure (miR-654-5p, miR-3197, and miR-6780b-5p), and one was nominally associated with PFNA (miR-210-5p). Furthermore, one EV-miRNA in Cluster 1 was associated with PFNA in individual models (miR-128-3p). KEGG pathway analysis for these EV-miRNAs indicated enrichment for Mucin type-O-glycan biosynthesis and morphine addiction for PFNA and PFOS, respectively (Supplemental Figs. S9 and 10). Following pathway analyses in TargetScan and Tarbase also revealed Mucin type-O-glycan biosynthesis as the top enriched pathway for PFNA and N-Glucan biosynthesis for PFOS (Supplemental Tables S5 and S6).

3.5. Sensitivity analyses

Our results were not meaningfully altered by adjustment for fish consumption in a subset of the participants ($n = 168$) (Supplemental Fig. S11), adjustment of models for gestational diabetes (Supplemental Fig. S12) or maternal BMI and gestational weight gain (Supplemental Fig. S13), exclusion of women with preeclampsia ($n = 4$) (Supplemental Fig. S14), or alternate library normalization with TMM (Supplemental

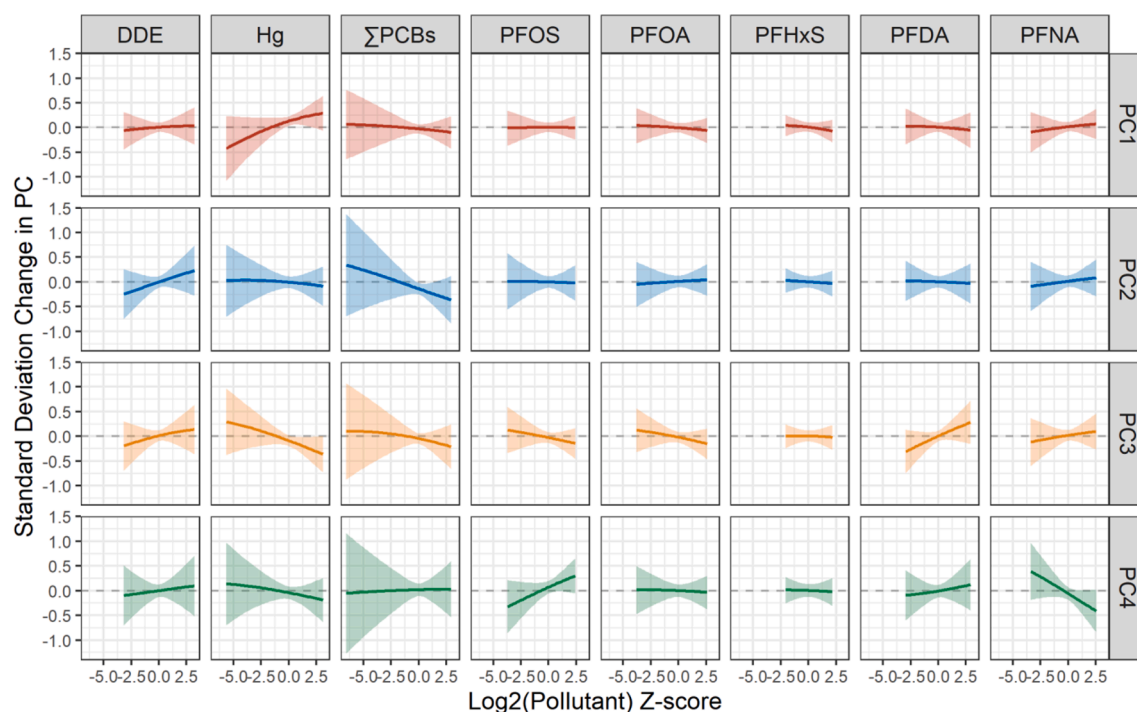


Fig. 2. BKMR univariate exposure-response plots with the 95% credible intervals for the effect each pollutant on PCs 1 – 4 when other pollutants are fixed at their median. All models adjusted for age at parturition, parity, smoking during pregnancy, education, RNA concentration, and milk collection day postpartum.

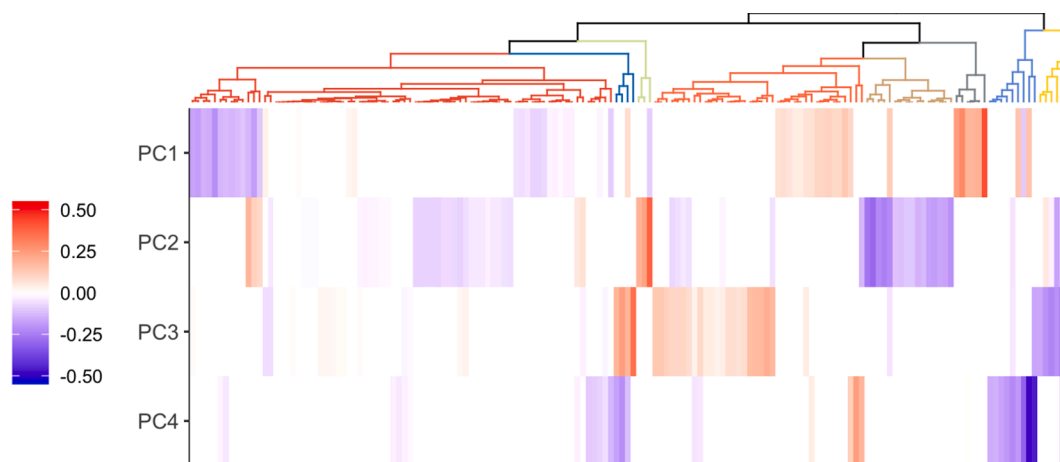


Fig. 3. Heatmap of EV-miRNA SPC loadings of 158 EV-miRNAs grouped with hierarchical clustering. Color gradients represent the positive (red) and negative (blue) loadings of individual EV-miRNAs. White suggests a loading shrunk to zero. Dendrogram colors denote the 8 groupings with clusters 1 – 8 arranged from left to right. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

[Fig. S15](#)). Adjusting models for child sex as a potential precision variable increased the FDR significance of the association between Hg and miR-128-3p but resulted in no other meaningful changes ([Supplemental Fig. S16](#)). Specifying models without confounders and adjusting for only EV-miRNA sequencing precision variables resulted in FDR significant associations between Hg and miR-128-3p (adjusted $p = 0.06$) ([Supplemental Fig. S17](#)). With the inclusion of potential influential points, we additionally observed an FDR significant association between PFOA and miR-4507 (Fold change = 0.24; adjusted $p = 0.04$) and between ΣPCBs and miR-128-3p (Fold change = 0.16; adjusted $p = 0.07$) ([Supplemental Fig. S18](#)). When the study population was limited to nonsmokers ($n = 242$), associations between pollutants and milk EV-miRNAs were unchanged for DDE, PFHxS, and ΣPCBs ([Supplemental Fig. S19](#)). However, associations between Hg, PFOA, and PFDA and miRNAs were attenuated, while several positive associations between PFNA and miRNAs

were observed; however, none of these associations reached statistical significance. When associations were examined in smokers ($n = 91$), 3 miRNAs were strongly associated with Hg concentrations at Bonferroni significance (miR-1290, miR-1291, and miR-6516-3p; Fold change = 1.48, 1.33, and 1.28, respectively) and 12 miRNAs were negatively associated with PFNA at FDR < 0.1 ([Supplemental Fig. S20](#)). Increasing or decreasing the L1 (lasso) penalty parameter in SPCA analyses did not greatly impact our results, except associations between ΣPCBs and PC2 were attenuated and Hg was now positively associated with PC2 ([Supplemental Fig. S21](#)). Finally, altering the BKMR prior distribution hyperparameters did not change our results ([Supplemental Figs. S22 and S23](#)).

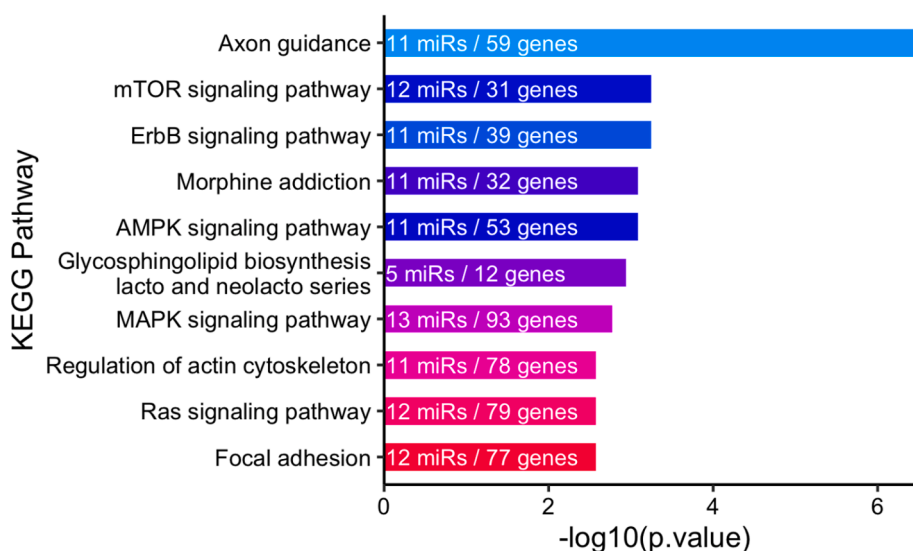


Fig. 4. Top 10 non-cancer related KEGG Pathways for 13 EV-miRNA mRNA targets in PC1 that are also nominally associated with Hg in individual models ($p < 0.05$).

4. Discussion

In the present study, we observed no associations between single marine pollutant exposures and individual human milk EV-miRNAs at $FDR < 0.1$. However, using dimensionality reduction and advanced mixture methods, we identified sets of biologically relevant milk EV-miRNAs that may serve as biomarkers of prenatal pollutant exposures.

EVs in milk are thought to be secreted primarily by mammary epithelial cells (Alsaweed et al., 2016b) and exposures to marine pollutants during pregnancy may influence milk EV-miRNA content via several mechanisms. During pregnancy, the breast undergoes numerous physiologic and anatomical changes that are characterized by decreased breast adipose tissue, expanded ductal tissue, and increased lobular branching (Alex et al., 2020). Environmental toxicants may induce systemic responses that interfere with mammary gland development and these processes, leading to cell stress and impeding normal physiology. Additionally, colostrum production begins during the 2nd trimester of pregnancy. Many environmental toxicants, including those evaluated herein, are known to partition into human milk due to their lipophilicity and protein binding (Grandjean et al., 1995, 1994; Mogensen et al., 2015). Thus, toxicants may directly reach mammary epithelial cells to induce specific stress responses that lead to altered miRNA expression and EV excretion.

We identified 13 EV-miRNAs that were both nominally associated with Hg in individual models and had non-zero loadings in PC1. Pathway analyses revealed that axon guidance was the most significantly enriched biological pathway associated with these EV-miRNAs, which was supported by using a second mRNA target prediction software. Methylmercury (MeHg) is a well-established developmental neurotoxicant, with established associations with neurodevelopmental deficits in this cohort (Debes et al., 2016; Oulhote et al., 2019). MeHg also has known impacts on axonal morphogenesis and differentiation in experimental models (Aaseth et al., 2020; Heidemann et al., 2001), through processes that may be captured by this pathway. Furthermore, mTOR signaling, the second most highly enriched pathway for these miRNA-mRNA targets, plays a critical role in cell metabolism, growth, and survival (Saxton and Sabatini, 2017), with a particular function in neuronal development (Switon et al., 2017; Takei and Nawa, 2014). Experimental research suggests that MeHg exposure dysregulates mTOR signaling in the brain (Heimfarth et al., 2018; Rosa-Silva et al., 2020), potentially eliciting neuronal cell death via mTOR-dependent autophagy (Ferrer et al., 2021; Lin et al., 2019, p. 201). While these EV-

miRNA may not reach the child brain following breastfeeding, the identification of this pathway suggests that EV-miRNAs in human milk may serve as mechanistic biomarkers of pollutant effects.

Additionally, individual EV-miRNAs associated with Hg within PC1 have also been demonstrated to be associated with Hg-relevant pathways. For instance, miR-429 and miR-200b-3p are part of the miR-200 gene family of miRNAs, which are induced during postnatal neurogenesis *in vivo* (Beclin et al., 2016), are neuroprotective *in vivo* and *in vitro* following hypoxia/reperfusion (Wei et al., 2015; Yang et al., 2020) and were upregulated in mesenchymal stem cells EVs following oxidative stress (Luo et al., 2021). miR-664-3p was also found to be differentially expressed in patients with Alzheimer's disease and major depressive disorder (Mendes-Silva et al., 2016). Other miRNAs in this cluster may be associated with diabetes (Kim et al., 2019; Ying et al., 2014).

We also observed associations between Σ PCBs and EV-miRNA PC2, with three miRNAs shared between PC2 and associations between Σ PCBs and individual EV-miRNAs. These miRNAs were predicted to impact thyroid hormone synthesis with thyroglobulin as a putative miR-1913 target. Human population literature suggests that PCBs may act as thyroid hormone disruptors, though findings are sometimes inconsistent (Benson et al., 2018; Hisada et al., 2013; Julvez et al., 2011). Thyroid hormone synthesis was also a significantly enriched miRNAs in PC3 that were associated with prenatal hair Hg concentrations. However, repeating pathway analyses using two different target prediction models did identify thyroid hormone signaling as a potential pathway, suggesting that these results should be interpreted with caution. Finally, PFOS and PFNA were associated with PC4 with mucin type O-glycan biosynthesis and morphine addiction as the top enriched pathways, respectively. However, the implications of these results are unclear and require further investigation.

To our knowledge, this is the first analysis examining the associations between environmental pollutant exposures and milk EV-miRNAs. However, select studies have investigated associations between environmental contaminants and miRNA expression in blood serum or plasma from human populations. For example, researchers investigating associations between high PFOA exposures and serum miRNAs amongst occupational workers identified nine differentially expressed miRNAs (Wang et al., 2012). Furthermore, a comparison of serum miRNA expression between PFAS exposed and unexposed individuals through drinking water identified three high-confidence miRNAs associated with PFAS exposure (miR-101-3p, miR-144-3p, and miR-19a-3p) (Xu et al.,

2020). However, none of these miRNAs were replicated in our study. An investigation of blood cell miRNAs collected from 207 Swedish adults found 93 miRNAs associated with summed concentrations of DDE, PCBs, and hexachlorobenzene (Krauskopf et al., 2017). One miRNA, miR-1304-5p, was also nominally associated with DDE in the present analysis, while miR-4135 was associated with Σ PCBs in the present analysis. A separate case-control study of female workers from a Hg thermometer plant in China found eight plasma miRNAs altered between control groups and chronic Hg poisoning groups (Ding et al., 2016). However, none of these miRNAs overlapped with those that were statistically significant in the present analysis. Discrepancies between our findings and those reported previously may be due to differing impacts of these exposures on blood and mammary epithelial cells or alterations in physiology surrounding birth and lactation. Additionally, the majority of the previous studies were conducted in occupational cohorts with different routes of exposure and much greater exposure concentrations than those quantified in the current study population.

The concentrations of Hg in this study were greater than recommended by regulatory bodies, with 40% of women having >2.5 $\mu\text{g/g}$ hair Hg (Bellanger et al., 2013). Furthermore, serum PCB and DDE levels were greater than those measured in more recent cohorts from the Faroe Islands and Europe (Karlsen et al., 2017; Vafeiadi et al., 2015). PFOS, PFOA, and PFHxS levels were greater than those measured in a more recent cohort from the Faroe Islands and those from North America, while PFNA and PFDA levels were similar (Karlsen et al., 2017; Vuong et al., 2021).

This study is the first of its kind to examine associations between milk EV-miRNAs and maternal pollutant exposures. The study has several strengths, including the comprehensive assessment of milk EV-miRNAs, application of advanced mixture and dimensionality reduction methods, analysis of data from a fairly homogenous cohort, and the high variability in pollutant exposures. However, this work has several limitations. As this study was performed in a small North Atlantic fishing community, these results may not be generalizable to other populations. The use of a targeted library preparation method, which while removing certain biases in miRNA sequencing, precluded the analysis of novel miRNAs or changes in miRNA processing. As with any human population study, there remains the potential for unmeasured or residual confounding. Although we were able to adjust for many important confounders, additional negative confounders, such as micronutrients, omega-3 fatty acids, or other dietary factors, may influence our results. Milk composition can vary depending on feeding or expression method, which we were unable to assess in our study. These factors may increase the uncertainty of EV-miRNA measurement and are expected to be nondifferential by exposure, which would generally lead to an attenuation of the observed associations and may be responsible for some of our null findings. Further, impacts on child development caused by milk miRNAs likely depend on the duration of breastfeeding and any changes in miRNA concentrations over time; these aspects were not considered in the present study. Finally, targets of miRNAs can be difficult to predict and results from pathway analyses must be interpreted with caution.

In sum, although no individual milk EV-miRNAs were associated with the marine pollutant measurements, we identified sets of pollutant-linked EV-miRNAs that may be useful as exposure biomarkers during the highly vulnerable infancy period. Further research would benefit from larger sample sizes and longitudinal analysis of milk EV-miRNAs. Future work should investigate associations between milk EV-miRNAs and infant health, as well as the role of maternal lactation health, such as mastitis, on milk EV-miRNA secretion.

CRedit authorship contribution statement

Allison Kupsco: Conceptualization, Investigation, Software, Formal analysis, Writing – original draft, Visualization. **Jenny Jyoung Lee:** Conceptualization, Methodology, Software, Writing – review & editing. **Diddier Prada:** Conceptualization, Writing – review & editing, Funding

acquisition. **Damaskini Valvi:** Conceptualization, Writing – review & editing. **Lisa Hu:** Investigation, Writing – review & editing. **Maria Skaalum Petersen:** Conceptualization, Resources, Data curation, Writing – review & editing. **Brent A. Coull:** Conceptualization, Methodology, Writing – review & editing. **Pal Weihe:** Conceptualization, Resources, Data curation, Funding acquisition, Supervision, Writing – review & editing. **Philippe Grandjean:** Conceptualization, Resources, Data curation, Funding acquisition, Supervision, Writing – review & editing. **Andrea A. Baccarelli:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2021.106986>.

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